

Grasp



Grade 4 (2025-2026)

SC4S2U2: Waves and Information Performance Task

Goal

You will build and show a model that uses light or sound waves to send a message.

You will also explain:

1. how waves move,
2. how light helps us see, and
3. how your model transfers information using patterns (like flashes or beats)

Role

You are a young engineer who designs new ways for people to communicate without phones or electricity.

Audience

Your classmates and teachers will watch your **Waves Communication Show**, where each group shows how their model works.

Situation

Sometimes people cannot use phones during storms or power cuts. Engineers have to find other ways to send messages, using light (like flashlights or mirrors) or sound (like drums or string phones).

Your job is to design a simple model that can send a message from one place to another using light or sound patterns.

Product/Performance & Purpose

You will:

1. Build a model that uses light or sound to send a short message (for example, "Help," "Go," or "Stop").
2. Show a wave model with amplitude (height) and wavelength (length between waves).
3. Explain or show how your model works:
 - How do the waves move?



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- How does light help people see the message?
 - What pattern do you use to send information?
4. Compare your design with another group's idea and say which one works better and why.

Standards and Criteria for Success

Science Understanding

- Show and label a wave model (amplitude and wavelength).
- Explain that light reflects from objects and goes into our eyes so we can see them.
- Use patterns in light or sound to send a clear message.
- Compare two designs and tell which one meets the goals best (clear signal, easy to build, safe).

Engineering & Communication

- Use clear labels and pictures to show how your model works.
- Use science words: wave, vibration, reflection, signal, pattern.
- Speak clearly and work well with your group during your presentation.



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Rubric

_____/12

Standard: 4-PS4-1. Develop a model of waves to describe patterns in terms of amplitude and wavelength and that waves can cause objects to move.

Exceeding	Meeting	Approaching	Emerging	Not Met
Developed a detailed and accurate model of waves to clearly describe patterns in terms of amplitude and wavelength and that waves can cause objects to move.	Developed an adequate model of waves to describe patterns in terms of amplitude and wavelength and that waves can cause objects to move.	Developed a partially accurate model of waves to describe patterns in terms of amplitude and wavelength and that waves can cause objects to move. The description may be lacking and/or include inaccuracies.	Struggled to develop a model of waves to describe patterns in terms of amplitude and wavelength and that waves can cause objects to move. The description displays significant misconceptions.	
Glossary				

Standard: 4-PS4-3.

Generate and compare multiple solutions that use patterns to transfer information.*

Exceeding	Meeting	Approaching	Emerging	Not Met
Generated and compared multiple solutions that use accurate patterns to transfer information.*	Generated and compared multiple solutions that adequately use patterns to transfer information.*	Generated and compared multiple solutions that partially use patterns to transfer information.* Comparisons are lacking and/or inaccurate.	Struggled to generate and compare multiple solutions that use patterns to transfer information.* Comparisons may include misconceptions.	
Glossary				



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Standard: 4-PS4-2.

Develop a model to describe that light reflecting from objects and entering the eye allows objects to be seen.

Exceeding	Meeting	Approaching	Emerging	Not Met
Developed a detailed and accurate model to clearly describe that light reflecting from objects and entering the eye allows objects to be seen.	Developed an adequate model to describe that light reflecting from objects and entering the eye allows objects to be seen.	Developed a partially accurate model to describe that light reflecting from objects and entering the eye allows objects to be seen. The description may be lacking and/or include inaccuracies.	Struggled to develop a model to describe that light reflecting from objects and entering the eye allows objects to be seen. The description displays significant misconceptions.	
Glossary				

Standard: 3-5-ETS1-2. Engineering Design

Generate and compare multiple possible solutions to a problem based on how well each is likely to meet the criteria and constraints of the problem.

Exceeding	Meeting	Approaching	Emerging	Not Met
Generated clear and detailed solutions and accurately evaluated how each met criteria and constraints by analyzing trade-offs and using a systematic process aligned with data or prior testing.	Adequately generated and compared multiple possible solutions based on a clear understanding of how well each was likely to meet the criteria and constraints of the problem.	Generated some solutions, but comparison or reference to criteria and constraints was partial or unclear.	Struggled to generate or compare solutions; understanding of criteria and constraints was incomplete or incorrect.	



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**SC4S2U2: Waves and Information
Performance Task**

Glossary

Student Handout T3



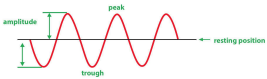
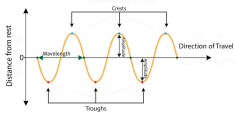
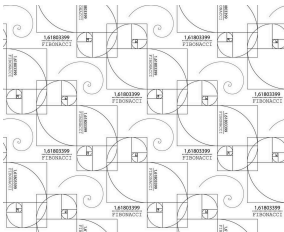
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SC4S2U2: Waves and Information

Tier 3 Performance Task

Visual Vocabulary & Planning Guide

Use this guide to help you plan your communication model. Review the vocabulary words and then choose how you will send your message.

Word & Image	Simple Definition
	Amplitude: The height of a wave. It shows how much energy the wave has.
	Wavelength: The distance between one wave peak and the next one.
	Pattern: A sequence that repeats over and over again.
	Signal: A message sent by using light or sound.

Step 1: Criteria and Constraints

Before you build, look at what your model **MUST do** and **the rules you MUST follow**. This helps you plan.

Criteria (What must it do?)	Constraints (What are the limits?)
Must send 3 different messages: HELP, GO, STOP.	Must use LIGHT waves only.
The receiver must read the pattern correctly.	Must use a repeating sequence (pattern).

Step 2: Choose Your Tool

How will you send your message? Circle one below:

 LIGHT	 MIRROR
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Step 3: Choose Your Message

Which short message will your model send? Circle one below:

 HELP	 GO	 STOP
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Step 4: Plan Your Message

Write your chosen message in the box below. Then, describe the pattern you will use (for example: 2 short flashes, or 1 long beep).

My Message and Pattern Plan:

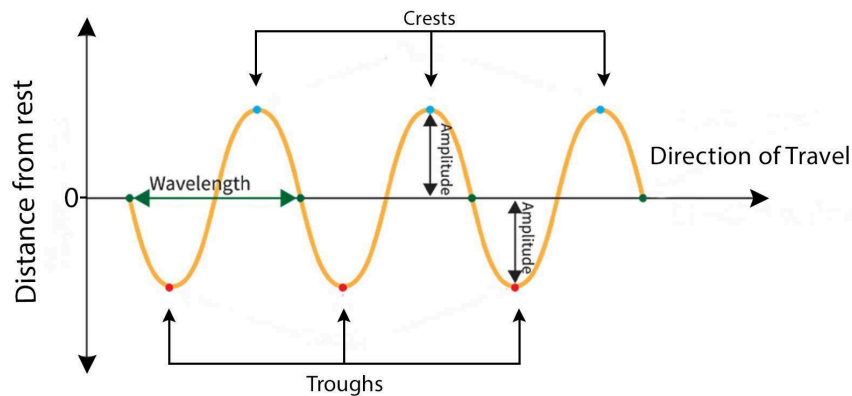
Step 5: Wave Model Diagram



Remember

- **Amplitude:** The height of the wave from the center line to the top (crest).
- **Wavelength:** The distance from the top of one wave to the top of the next wave.

Directions: Look at the wave diagram below. Use the empty box to draw the specific waves your model uses to send your message.



Example of a labeled wave diagram

Draw your wave model here:

Explain Your Wave Design

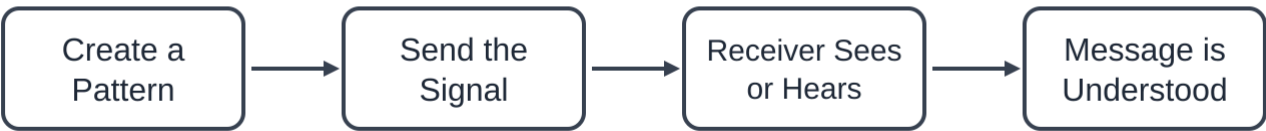
My wave has a **high / low** (circle one) amplitude because:

My wavelength is **long / short** (circle one) because:

Design Explanation & Pattern Key

Now that you have planned your signal, it is time to explain how your model works. Follow the steps below to finish your design guide.

How Information Travels



This flow map shows how your message moves from you to your friend.

✓ Step 1: Function Checklist

Check off each box as you test your model:

☐ My model uses a clear **Pattern** (repeating sequence).

☐ My model sends a **Signal** using light or sound.

☐ My model can send the message **HELP**, **GO**, or **STOP**.

☐ A person far away can see or hear the message clearly.

✓ Step 2: Information Pattern Key

What the Receiver Sees	What the Message Means
	SOS HELP
	GO
	STOP

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✓ Step 3: The Science of Your Design	
Use the sentence frames below to explain your work:	
The light helps people receive the message by...	
	The pattern I used to send information is...

Design Comparison & Evaluation

Now that you have built your wave model, it is time to look at your work and compare it with another. Scientists always check their designs to see how they can make them better! 🚀

 Tool 1	 Tool 2
Which light tool?	Which light tool?
How does it meet the criteria?	How does it meet the criteria?
How does it handle the constraints?	How does it handle the constraints?

 Success Checklist	
Was the message (Help, Go, or Stop) clear?	Yes / No
Did the pattern repeat correctly?	Yes / No
Could the receiver see or hear the signal from far away?	Yes / No

Final Evaluation:

I think (My Design/ Other's) design worked better because...

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Criteria and Constraints: Evaluating Solutions

Engineering Definitions

Criteria: The requirements for success. These are the goals that the solution must achieve to be considered successful (e.g., "The message must be understood from 20 feet away").

Constraints: The limits or rules that you must follow. These are often restrictions on time, money, or materials (e.g., "You can only use the materials in this bin").

Wave Communication Task: Requirements and Limits	
Criteria (What must it do?)	Constraints (What are the limits?)
• Must send 3 different messages (HELP, GO, STOP).	• Must only use light or sound waves.
• The receiver must be able to decode the pattern correctly.	• Must use a repeating sequence (pattern).

Solution Comparison

Compare two different tools or patterns for sending your signal. For example, you might compare using a **Flashlight** versus a **Mirror**.

Solution Name	How it meets Criteria	How it handles Constraints

Based on your comparison, which solution is most likely to succeed and why?

The solution that will succeed is _____ because....

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